

Predictive vs. Descriptive vs. Prescriptive Analytics

Understanding the Three Main Types of Data Analytics



Descriptive: What happened?

Descriptive Analytics



Predictive: What could happen?

Predictive Analytics



Prescriptive: What should we do?

Prescriptive Analytics

How do we make it happen?

Descriptive Analytics

Understanding What Happened

? Definition

Analysis of **historical data** to understand past events and trends.

Answers the question: **"What happened?"**

🔧 Techniques

Data Aggregation

Data Mining

Data Visualization

Statistical Analysis

💻 Tools

Excel

Tableau

Power BI

SQL

📊 Applications

Business Reporting

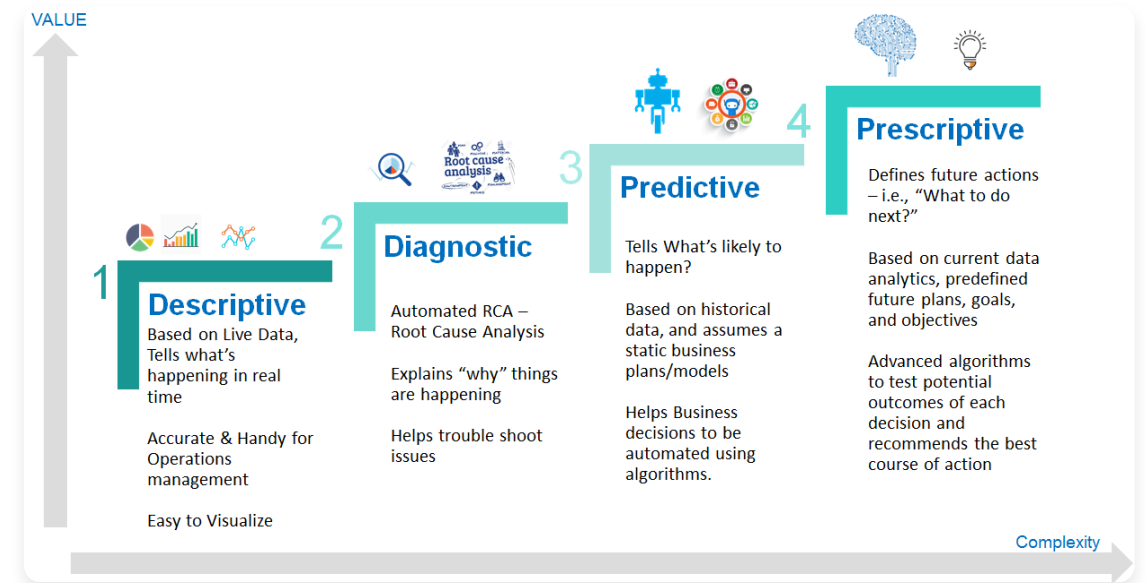
Customer Segmentation

Market Analysis

Operational Efficiency

Real-World Example

Annual revenue reports, social media analytics, and sales dashboards that summarize past performance



⚡ Key Characteristics

- Uses **historical data**
- Focuses on **past events**
- Provides **context** for future analysis
- Often displayed through **visual representations**

Predictive Analytics

Forecasting What Could Happen

? Definition

Uses **historical data** and **statistical algorithms** to forecast future events. Answers the question: "**What could happen?**"

🔧 Techniques

Regression Analysis

Time Series Analysis

Machine Learning

Classification

Clustering

💻 Tools

R

Python

SAS

IBM SPSS

🏢 Applications

Risk Management

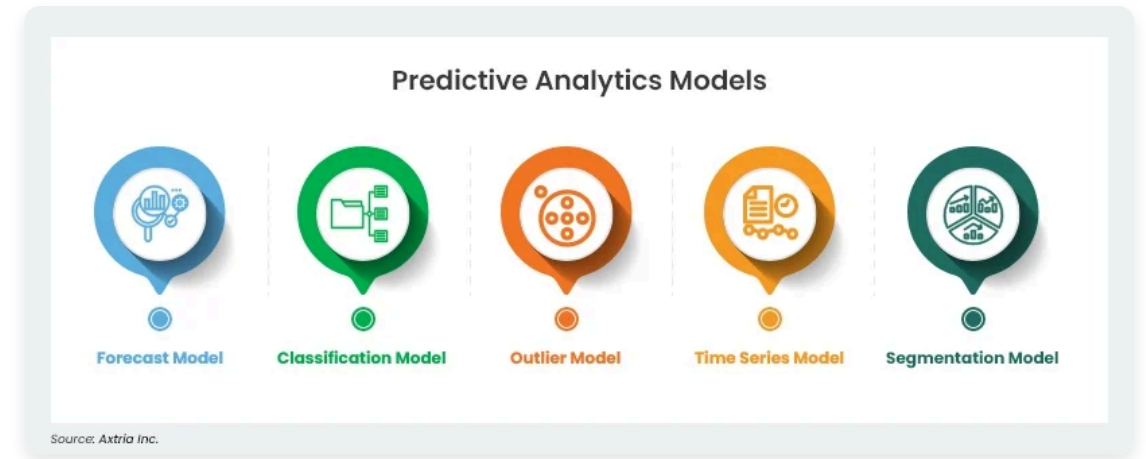
Customer Retention

Sales Forecasting

Healthcare

Real-World Example

E-commerce businesses using customer's browsing history to recommend products and predict future purchases



⚡ Key Characteristics

- Uses **statistical models**
- Focuses on **future outcomes**
- Enables **proactive decision-making**
- Requires **quality historical data**

Prescriptive Analytics

Recommending What Should We Do

🔗 Definition

Goes beyond prediction by **recommending actions** to achieve desired results. Answers the question: **"What should we do?"**

🔧 Techniques

Optimization

Simulation

Decision Analysis

Machine Learning

💻 Tools

Gurobi

IBM ILOG CPLEX

AnyLogic

MATLAB

🏢 Applications

Supply Chain Optimization

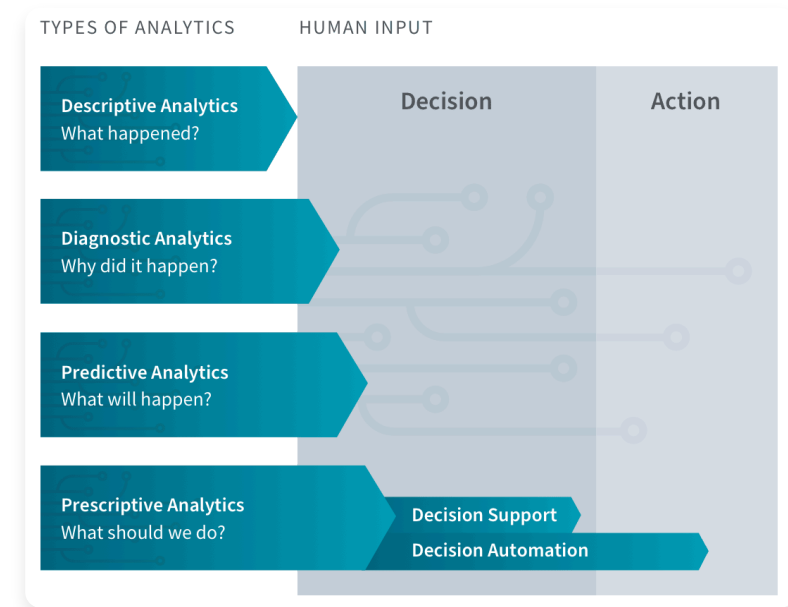
Revenue Management

Healthcare

Finance

Real-World Example

Identifying techniques to optimize patient care in healthcare by recommending personalized treatment plans



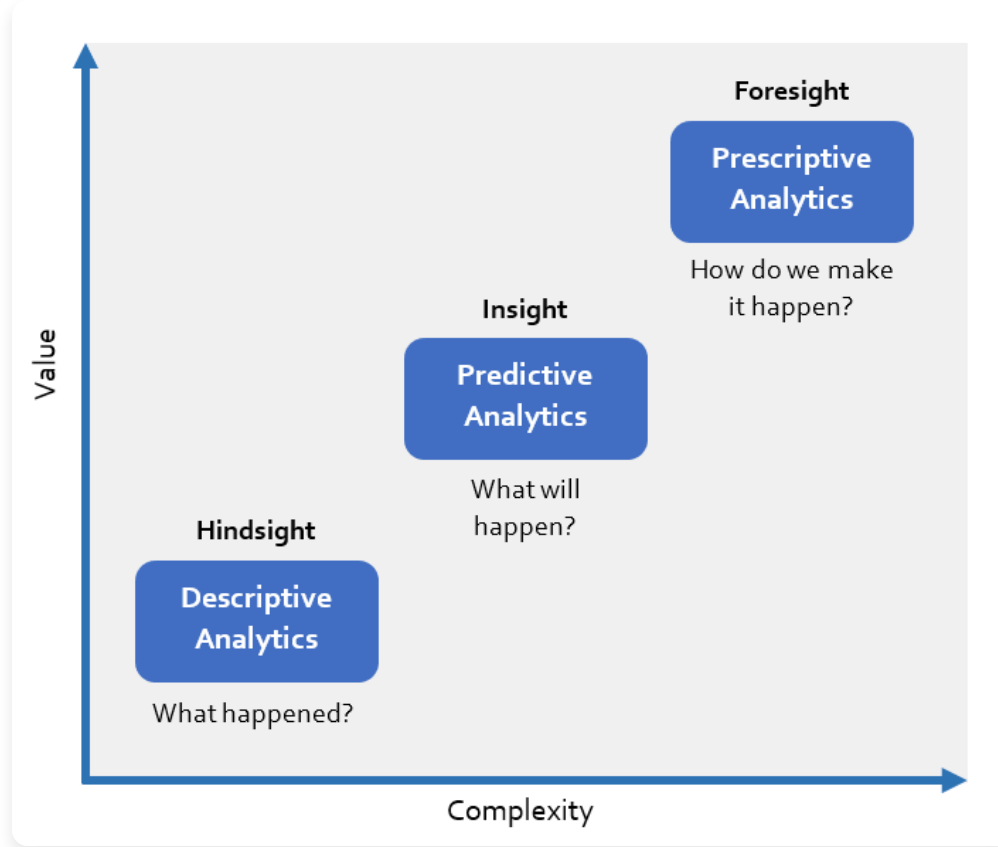
🌟 Key Characteristics

- Uses **optimization algorithms**
- Provides **actionable recommendations**
- Considers **multiple scenarios**
- Enables **optimal decision-making**

Comparison of Analytics Types

Key Differences Between Descriptive, Predictive, and Prescriptive Analytics

Aspect	Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Focus/Purpose	Understanding past events and trends	Forecasting future events	Recommending actions to achieve desired outcomes
Question Answered	"What happened?"	"What could happen?"	"What should we do?"
Analysis Approach	Analysis of historical data to understand trends	Analysis to determine future trends	Analysis showcasing feasible solutions and their impact
Tools Used	Data mining, Data aggregation	Machine learning, Statistical models	Heuristics, Optimization
Approach	Reactive	Proactive	Proactive
Examples	Annual Revenue Report	E-commerce product recommendations	Healthcare treatment optimization



Analytics Strategy: Working Together

How the Three Analytics Types Create a Complete Solution

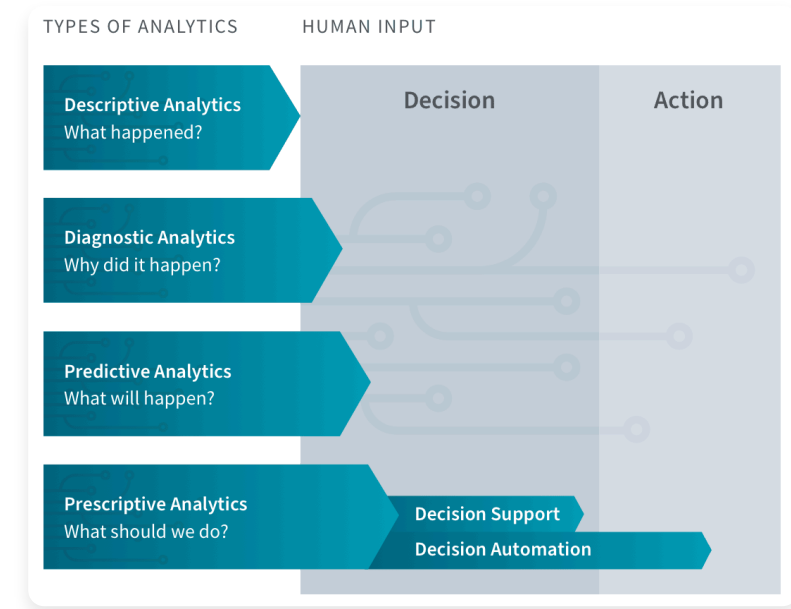
Integrated Analytics Approach

The three analytics types form a **progressive framework** that builds upon each other to deliver maximum value:



★ Benefits of Integrated Analytics

- ✓ Comprehensive insights
- ✓ Improved decision-making
- ✓ Competitive advantage
- ✓ Increased ROI



Strategic Implementation

- Start with **descriptive analytics** to establish baseline
- Add **predictive capabilities** to forecast trends
- Implement **prescriptive analytics** for optimization
- Continuously refine models with new data

Real-World Examples & Case Studies

How Organizations Leverage Different Analytics Types



Retail

Descriptive

Predictive

Prescriptive

- **Descriptive:** Sales trend analysis
- **Predictive:** Customer purchase forecasting
- **Prescriptive:** Dynamic pricing optimization

Key Result

Amazon increased revenue by 35% using prescriptive analytics for personalized recommendations



Healthcare

Descriptive

Predictive

Prescriptive

- **Descriptive:** Patient history analysis
- **Predictive:** Disease risk assessment
- **Prescriptive:** Treatment plan optimization

Key Result

Mayo Clinic reduced hospital readmissions by 20% using predictive analytics



Finance

Descriptive

Predictive

Prescriptive

- **Descriptive:** Transaction pattern analysis
- **Predictive:** Credit risk scoring
- **Prescriptive:** Portfolio optimization

Key Result

Mastercard prevented \$25B in fraud using predictive analytics



Manufacturing

Descriptive

Predictive

Prescriptive

- **Descriptive:** Production performance metrics
- **Predictive:** Equipment failure prediction
- **Prescriptive:** Supply chain optimization

Key Result

GE reduced maintenance costs by 30% using prescriptive analytics

Conclusion & Key Takeaways

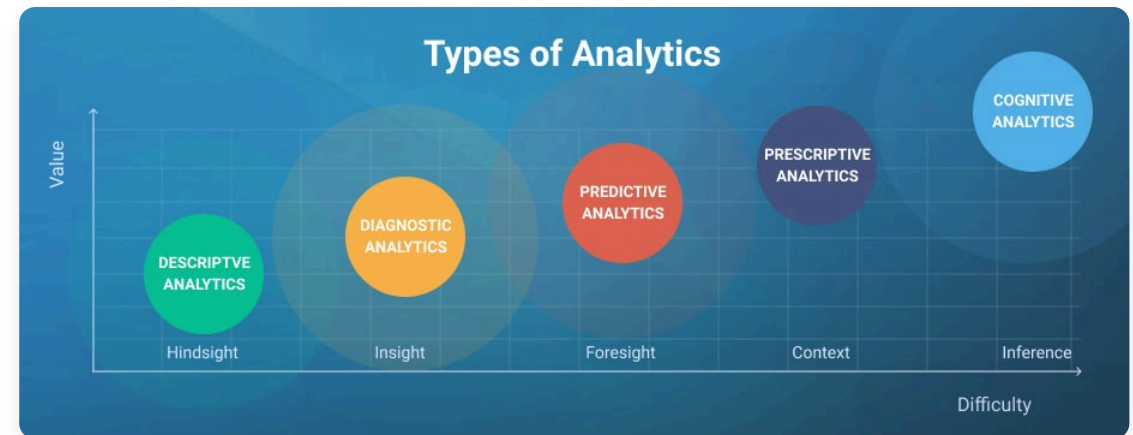
Leveraging Analytics for Data-Driven Decision Making

💡 Key Takeaways

- 🕒 **Descriptive Analytics** provides the foundation by answering "What happened?"
- 📈 **Predictive Analytics** forecasts future outcomes by answering "What could happen?"
- 💡 **Prescriptive Analytics** recommends actions by answering "What should we do?"
- 🔗 The **greatest value** comes from integrating all three types into a comprehensive analytics strategy

🧠 Final Thoughts

Organizations that effectively implement all three analytics types gain significant competitive advantage through **data-driven decision making**, improved operational efficiency, and enhanced customer experiences.



📈 Future Trends in Analytics

- ✦ AI and machine learning integration
- 🔄 Real-time analytics capabilities
- 👥 Democratization of analytics tools
- 🛡️ Enhanced focus on data privacy and ethics